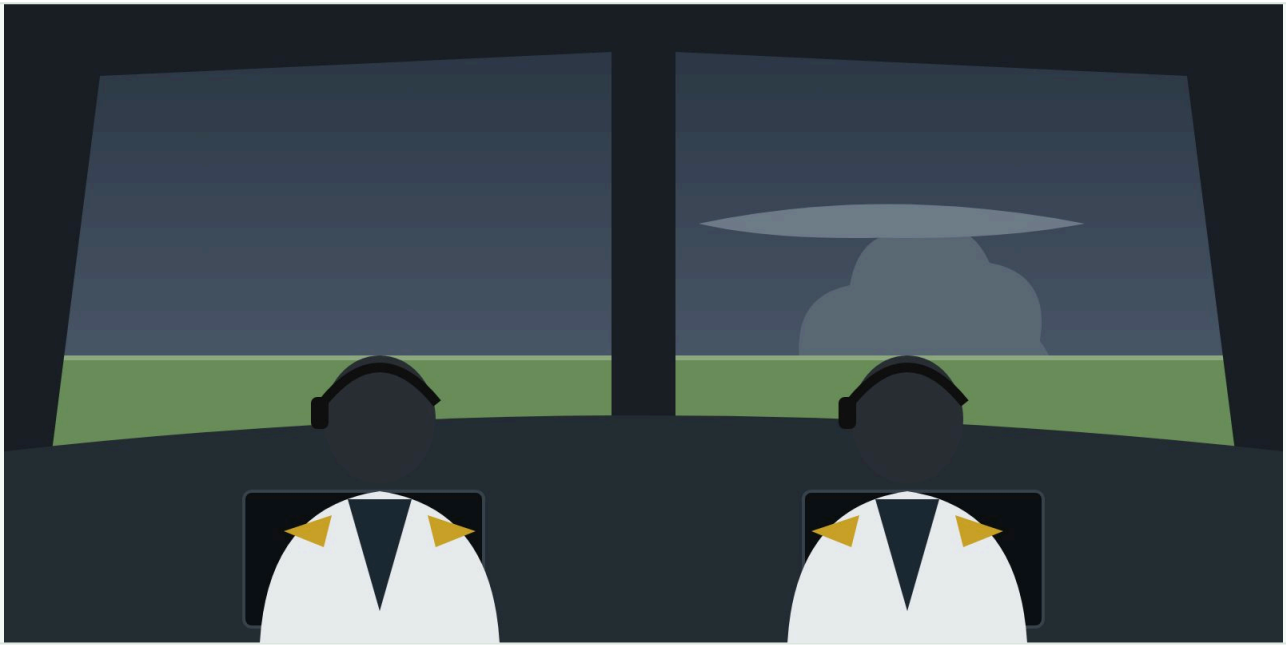
MODULE 2 · THUNDERSTORMS AND PLANNING

Lesson 05: Planning, CRM and Lessons Learned

Lesson 5 of 5 · about 6 min

LEARNING OBJECTIVES

- Integrate windshear and thunderstorm threats into planning and briefings.
- Apply decision points and CRM.
- Learn from a classic windshear accident.



Brief the threat and agree the decision points.

Planning and briefing

- Brief convective weather on the route and at destination, alternates, extra fuel for deviations and holding.
- Before take-off or approach in convective weather: brief the windshear escape, the go-around and the decision points (“if windshear is reported on final, we hold / divert”).
- Agree on the callouts for speed and vertical speed deviations.

CRM

- Plan continuation bias is a strong factor in convective weather — “the aircraft ahead landed”. Each aircraft meets a different part of the storm.
- Either pilot can call for the escape manoeuvre; the PF executes immediately.
- Use ATC and other crews' reports, and give your own.

Lesson learned — Delta Air Lines 191 (Dallas/Fort Worth, August 1985)

A Lockheed L-1011 on approach flew into a microburst from a thunderstorm close to the runway. The aircraft lost airspeed and height and struck the ground short of the runway; 137 people died. The investigation led directly to windshear training programmes and to the development of airborne windshear detection systems.

KEY MESSAGE

Do not penetrate a thunderstorm or approach through one. Waiting 20 minutes for a storm to pass costs fuel — never safety.

KEY TAKEAWAYS

- Brief convective threats, the escape manoeuvre and decision points.
- Beware of plan continuation — each aircraft meets a different storm.
- Delta 191 (1985) shaped modern windshear training and detection systems.

KNOWLEDGE CHECK

An aircraft ahead landed without problems in a thunderstorm area. What does this mean for you?

A. It is safe to land

B. Nothing — conditions in a storm change within minutes; apply your decision points

C. You should land faster

D. You may ignore windshear warnings

Answer: B. Each aircraft meets a different part of a rapidly changing storm.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.